

PSM Temp Logger

Temperature-dependent LCR and I-V characterisation suite
for the Newtons4th PSM1735, DPI-1100 furnace and Keithley 2400-series

User Manual

Developed by
MOHD VASIM, HIBA KHAN and SOHUM BISWAS

Acknowledgement

This project - the complete automation of the Newtons4th PSM1735 NumetriQ LCR meter, integration with the Divya DPI-1100 dry-block temperature calibrator and the Keithley 2400-series source meter, and the development of a single host application that drives all three for temperature-dependent measurements - was designed and brought to completion by **Mohd Vasim, PhD**, together with his interns **Hiba Khan** and **Sohum Biswas**.

Their work combined hardware bring-up (instrument communication, RS-232 / RS-485 / Modbus RTU / VISA / Arduino firmware) with the design and implementation of a unified Python host application capable of running unattended experiments across the full range of -273 to +1000 deg C, capturing LCR and I-V responses at every setpoint and writing the results directly to an Excel workbook for downstream analysis.

The authors are gratefully acknowledged for the time, persistence and care that went into the project - without which the workflow described in this manual would not exist.

1. Overview

The PSM Temp Logger automates electrical characterisation across user-defined temperature ranges. At each setpoint the application can run an LCR frequency sweep on the PSM1735, an I-V sweep on a Keithley 2400-series source meter, or both - with the temperature either held by a small Arduino-driven relay heater or by a Divya DPI-1100 dry-block calibrator over Modbus RTU. All captured data is written to an Excel workbook (one sheet per setpoint) and displayed live in the application.

Key features

- Bidirectional control of the PSM1735 over USB-serial (N4L ASCII).
- PT100 temperature reading via MAX31865 + Arduino, OR via the BTC-9100 PID controller inside a DPI-1100 calibrator over Modbus RTU.
- Bang-bang heater control with hysteresis, host watchdog, and a hard safety cut-off configurable at run-time.
- Keithley 2400 / 2401 / 2410 / 2420 source-meter support over VISA (USB-TMC, GPIB, Ethernet) or RS-232.
- Three measurement modes: LCR sweep only, I-V sweep only, or combined LCR + I-V at each setpoint.
- Live Bode and Nyquist plots (colour-coded by temperature) and tabular sweep viewer with CSV export.
- Real-time single-frequency LCR readout for sample alignment.
- Excel output with one sheet per setpoint, ready for post-processing.

2. Hardware setup

The full system can use any subset of the following instruments - the application requires only the ones relevant to the experiment you are running. PSM is required for LCR work; Keithley for I-V; Arduino or DPI-1100 (or neither, for room-temperature only) for the heat source.

Instrument	Connects via	Used for
Newtons4th PSM1735	USB (FTDI virtual COM)	LCR / impedance measurements
Arduino + MAX31865 + PT100 + relay	USB (CH340 / FT232 virtual COM)	PT100 temperature read; relay-switched heater control
Divya DPI-1100 (with internal BTC-9100)	RS-232 or RS-485 via USB adapter (Modbus RTU)	Dry-block temperature calibrator (range 50-650 deg C)
Keithley 2400 / 2401 / 2410	USB-TMC (VISA), GPIB, Ethernet, or RS-232	Source-measure unit for I-V characterisation

Arduino + heater pinout

Arduino pin	Connects to	Notes
D10	MAX31865 CS	SPI chip select
D11 / D12 / D13	MAX31865 SDI / SDO / SCK	Hardware SPI bus
5V / GND	MAX31865 and relay module VCC / GND	
D7	Relay module IN1	Polarity set by SSR_ACTIVE_HIGH in the firmware

Relay wiring

The heater is connected through the relay's **COM** and **NO** (Normally Open) terminals so that when the Arduino is unpowered the relay is de-energised and the heater is OFF. Do NOT use NC for the heater - that would leave the heater on if the controller dies.

DC supply (+) → relay COM → relay NO → heater (+) → heater (-) → DC supply (-).

Safety note: the firmware enforces a hard temperature ceiling (default 1050 deg C, overridden by the app per experiment) and a 60 s host watchdog. If the PC loses contact with the Arduino for more than 60 s the heater is automatically forced OFF.

3. Software installation

Required on the PC (one-time):

- **Python 3.9 or later** - download from python.org. Tick *Add Python to PATH* during install.
- **Python libraries** - open PowerShell or Command Prompt and run:

```
pip install pyserial openpyxl matplotlib pymodbus pyvisa pyvisa-py
```

- **Arduino IDE** - from arduino.cc. Install the *Adafruit MAX31865* library via *Tools* → *Manage Libraries*.
- **USB-serial drivers** - usually auto-installed by Windows. If an instrument does not appear as a COM port in Device Manager, install FTDI VCP drivers (for the PSM), CH340 drivers (for cheap Arduino / RS-485 adapters) or Prolific drivers (for PL2303 RS-232 adapters - install the older 3.3.2.105 driver for clone chips).
- **NI-VISA (optional)** - only needed if you are connecting a Keithley over GPIB. For USB-TMC and Ethernet the pure-Python pyvisa-py backend is sufficient.

4. Arduino firmware

Open `arduino\max31865_temp\max31865_temp.ino` in Arduino IDE, select the correct board (*Tools* → *Board*) and serial port (*Tools* → *Port*), then click **Upload**.

Edit the constants at the top of the sketch if your hardware differs (SSR_PIN for the relay control pin, SSR_ACTIVE_HIGH for the relay polarity, RTD_NOMINAL / REF_RESISTOR for PT100 vs PT1000, WIRES for the PT100 wiring, MAX_SAFE_TEMP_DEFAULT for the safety ceiling).

Serial commands the sketch accepts

Command	Argument	Effect
SET:<value>	Target deg C	Switch to AUTO mode, regulate to the specified value
LIMIT:<value>	Hard cut-off deg C	Change the safety ceiling at run time
H:1	(none)	Force heater ON (overrides target)
H:0	(none)	Force heater OFF
AUTO	(none)	Return to SET-driven control
PING	(none)	Heartbeat - resets the watchdog

Streaming output from the sketch

Line	Meaning	Example
T:<value>	Temperature in deg C, approx 5 Hz	T:27.482
H:<0 1>	Current heater state	H:1
ERR:<hex>	MAX31865 fault flags	ERR:84
#<text>	Diagnostic comment (ignored by host app)	# pong

5. DPI-1100 / BTC-9100 furnace

The Divya DPI-1100 dry-block temperature calibrator contains a Brainchild BTC-9100 PID controller. Communication is over Modbus RTU - either RS-232 or RS-485 depending on which optional interface module is fitted (CM97-2 = RS-232, CM97-1 = RS-485).

Default communication settings

Parameter on controller	Default value	Set in the app
COMM	RTU (= Modbus RTU)	(implicit)
ADDR	1	Slave
BAUD	9.6 (= 9600 bps)	Baud
DATA	8	(fixed: 8 data bits)
PARI	EVN (= Even parity)	Parity = E
STOP	1	(fixed: 1 stop bit)

Wiring (RS-232 with USB-RS232 adapter)

Controller terminal	DB9 pin on PC adapter	Function
Pin 13 (TXD)	Pin 2 (RD)	Controller transmits to PC
Pin 14 (RXD)	Pin 3 (TD)	PC transmits to controller
Pin 15 (COM)	Pin 5 (GND)	Common ground
-	Pin 1 to Pin 6 (jumper)	DCD-DSR loopback (handshake bypass)
-	Pin 7 to Pin 8 (jumper)	RTS-CTS loopback (handshake bypass)

Wiring (RS-485 with USB-RS485 adapter)

Controller terminal	Adapter terminal	Function
Pin 13 (TX1)	D+ (sometimes labelled A)	Differential pair, polarity not fixed - swap if no response
Pin 14 (TX2)	D- (sometimes labelled B)	Differential pair (other half)
Pin 15 (COM)	GND (optional)	Recommended for noise reduction

In the application, switch to the **Furnace (DPI-1100)** tab to connect to the controller and to read PV / write SP1 manually. Once connected, switch the Heat source selector on the Experiment tab to **DPI-1100 furnace** and the temperature loop will use the BTC-9100 as the thermal controller instead of the Arduino.

6. Keithley 2400-series source meter

The application supports the Keithley 2400 / 2401 / 2410 / 2420 / 2430 / 2440 source meter family. They share a common SCPI command set. Communication is via VISA (USB-TMC, GPIB, Ethernet) or directly via RS-232.

Connection options

Interface	VISA resource example	Notes
USB-TMC	USB0::0x05E6::0x2400::123456::INSTR	Click Scan VISA in the Keithley tab to auto-detect
GPIB	GPIB0::24::INSTR	Requires NI-VISA installed (pyvisa-py does not support GPIB)
Ethernet	TCPIP0::192.168.1.50::INSTR	Newer 2400 units with the LXI option
RS-232	(no VISA resource; pick COM port instead)	Use the Serial (RS-232) interface mode in the Keithley tab

Keithley tab features

Section	Purpose
Keithley connection	Pick Interface (VISA or Serial), select resource string or serial port, click Connect. *IDN? button verifies the link.
Manual source / measure	Configure source function (voltage or current), level and compliance, turn Output ON / OFF, take a single READ to display V and I.
IV sweep	Run a point-by-point sweep from Start to Stop over N points, with compliance and settle time. Each point is streamed to the log. Captured sweeps can be exported to CSV.

When the Keithley is connected, switch the Measurement selector on the Experiment tab to **IV sweep only** for standalone I-V characterisation, or **LCR + Keithley (IV) at each setpoint** to combine both instruments inside the temperature loop.

Safety note: always set a compliance limit appropriate for the device under test before turning the output on. Compliance is the safety limit on the *measured* quantity (current if sourcing voltage, voltage if sourcing current). The Keithley will clamp at this limit if the DUT impedance forces it to exceed.

7. Launching the application

From PowerShell:

```
cd "C:\Users\HIBA KHAN\Downloads\psm_temp_logger" python app\main.py
```

Or double-click run.bat. The application opens maximised with the following tabs:

Tab	Purpose
Experiment	Heat source and measurement type selectors, temperature range, Excel output path, Start / Stop / Capture controls, live status, heater manual override, message log.
PSM Configuration	All PSM1735 parameters organised by section (Sweep, OUT, CH1, CH2, AUX, MODE, ACQU, TRIM, SYS). Scrollable. Tooltips on every field. Sent to the instrument via the Apply settings button.
Live Graph	Matplotlib-embedded Bode or Nyquist plot. Updates after each captured sweep, colour-coded by temperature. Log / linear axes, show all sweeps or only the latest, abs value toggle.
Table	Tabular view of any captured sweep. Per-row CSV export of the entire dataset.
Realtime	Live single-frequency LCR readout updated approximately twice per second. Used for sample alignment and quick checks before starting a sweep.
Furnace (DPI-1100)	Connect to the BTC-9100 controller over Modbus RTU. Live PV, active setpoint, mode, alarm status, manual SP1 write, Reset and Auto-tune commands.
Keithley (2400)	Connect to a Keithley source meter over VISA or serial. Manual source / measure, IV sweep configuration and execution, CSV export.

8. Running an experiment

Step-by-step:

1. **Connect** - In the Experiment tab, click Refresh ports, pick the PSM port (default COM12) and (optionally) the Arduino port (default COM11), click Connect. If running an IV-only experiment you do not need to connect the Arduino.
2. **Configure instruments** - Switch to PSM Configuration and sanity-check Sweep / OUT / AUX / MODE for the LCR run. If using the Keithley or Furnace, also connect those in their respective tabs.
3. **Pick Heat source** on the Experiment tab: Arduino + relay, DPI-1100 furnace, or None (room temperature only).
4. **Pick Measurement**: LCR sweep only, IV sweep only, or LCR + Keithley (IV) at each setpoint.
5. **Pick experiment mode**: Stepped (manual Capture at each setpoint) or Continuous (auto-capture when measured temperature crosses each target within the tolerance band).
6. **Enter temperature range** - Start, End and Step. The application generates a list of setpoints from Start to End in increments of Step.
7. **Pick the Excel output file** - Browse to the destination. Parent folders are created automatically; a placeholder workbook is saved immediately so the file appears even before any sweep completes.
8. **Start experiment** - The application sends the configuration to all selected instruments, sets a safety LIMIT on the heater, and begins the loop. The Log panel streams progress.
9. **Monitor** - Switch to Live Graph, Table or Realtime tabs while the experiment runs. Heater state and live temperature are shown in the Experiment tab's status panel.
10. **Stop** - Either let it complete all setpoints, or click Stop. Stop also sends H:0 / SP1=min so the heater (or furnace) is brought to a safe state.

9. Output files

After each successful sweep the application appends a new worksheet to the Excel workbook. Sheet names are formatted as T_+25.0C, T_-50.0C, etc. for LCR data; combined LCR + I-V runs append a second sheet per setpoint with an _IV suffix.

Per-sheet layout (LCR)

Cell	Content	Example
A1, B1	Timestamp	2026-06-20 16:42:11
A2, B2	Target temperature (deg C)	+40.0
A3, B3	Measured temperature (deg C)	+40.12
A4 onwards	Experiment metadata	Amplitude, sweep range, mode, tolerance, direction
Header row	Column labels (8 columns)	Frequency_Hz, Q, TanD, Impedance_Ohm, Phase_deg, L_H, C_F, R_Ohm
Data rows	One row per frequency point	32 rows by default

CSV export from the Table tab writes the full dataset (all sweeps for all temperatures) into a single long-form CSV file with TargetTemp_C and MeasuredTemp_C columns prepended.

CSV export from the Keithley tab writes captured I-V sweeps with columns Sweep, Point, V, I, R.

10. Credits

Automation of the PSM1735 NumetriQ LCR meter, integration with the DPI-1100 / BTC-9100 dry-block calibrator, and Keithley 2400-series source-meter support - including all hardware integration, firmware, host application and documentation - completed by **Mohd Vasim, PhD** and his interns **Hiba Khan** and **Sohum Biswas**.

Reference documents: Newtons4th PSM1700 / PSM1735 Communications Manual (Rev 1.54), Brainchild Classic Premium 100 Series User's Manual UM91001I (chapter 7), Keithley 2400-series SCPI command reference.

Built on Python (Tkinter, pyserial, openpyxl, matplotlib, pymodbus, pyvisa, pyvisa-py) and the Adafruit MAX31865 Arduino library.

This manual was generated automatically from the project source. For the latest version of the code, sync the folder psm_temp_logger\.